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Henry's law constants of volatile organic compounds between 0 and 95 °C – Data compilation and complementation in context of urban temperature increases of the subsurface

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HIGHLIGHTS

- Compilation of Henry's law constants for 44 organic compounds as a function of temperature.
- Low availability of Henry's law constants at temperatures above 40 °C for most of the considered compounds.
- The temperature dependency of Henry's law constants of VOCs is strongly compound specific.

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ABSTRACT

In urban areas with frequently occurring contamination by volatile organic hydrocarbons (VOC) possible uncontrolled contaminant mobilization e.g. by volatilization is feared in case of subsurface temperature increases induced by high temperature underground thermal energy storage (HT-UTES) or due to urban heat islands (UHI).

However, volatilization of VOC is the main process utilized by thermal remediation approaches, wherefore a combination of contaminant remediation with UTES is increasingly discussed. To predict VOC volatilization under changing temperature conditions, which is substantially characterized by Henry's law constants (H), temperature dependency of H must be known substance-specifically in the total concerned temperature range.

To provide an overview of available H and to evaluate their quality, published data of 41 pollutants were compiled, revealing the need for further measurements above 40 °C for at least 24 compounds (as e.g. TCE/PCE, BTEX). Additionally, the temperature dependence of H was calculated based on the compiled data using an established temperature regression function. Furthermore, H was complementarily measured for 10 relevant VOCs between 10 and 90 °C using the EPICS-method (Equilibrium Partitioning In Closed Systems). The measurements quantified the expected strong increase in H with rising temperature while providing high data quality ($R^2 = 0.84$ – 0.99 , mostly low standard deviations), thus improving the general data availability of H for VOCs and extending the scope of the regression function. The newly measured data and enhanced evaluation of compiled data contribute to a more reliable assessment of the pollutant behaviour in terms of volatilization at elevated temperatures caused e.g. by UTES or UHI.

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1. Introduction

In urban areas with frequently occurring contamination by volatile organic hydrocarbons (VOC), anthropogenic temperature

increases of the subsurface by e.g. (high temperature) underground thermal energy storage (HT-UTES) with injection temperatures of up to 90 °C in shallow aquifers (Mangold et al., 2002; Sommer et al., 2013) or urban heat islands (UHI) where the aquifer temperature can increase by up to 20 °C (Menberg et al., 2013) are feared to cause spreading or relocation of the contamination. But it is also suggested to remediate contaminated sites in combination with UTES (Hoes et al., 2012; Ni et al., 2016; Ni et al., 2018; Picone et al.,

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2019; Sommer et al., 2013; VDI 4640, Part 3, 2001; Zuurbier et al., 2013). To predict or exclude an environmental hazard or estimate a remediation potential due to heating of contaminated sites, site-specific models are needed. Therefore, it is important to evaluate to which extent the current literature is sufficient to predict the effect of temperature increases in contaminated aquifers with regard to the complex transport, emission and transformation processes of pollutants. It has been shown that solubility (Koprock et al., 2019), dissolution kinetics (Imhoff et al., 1997; Koprock et al. in prep.), sorption equilibria (Bell and LeBoeuf, 2013), biodegradation (Ni et al., 2015), NAPL mobilization (Sleep and Ma, 1997) and gas phase formation (Hegele and Mumford, 2014; Heron et al., 1998), react directly to the ambient subsurface temperature. Especially the processes of gas phase formation need further investigation, as volatilization can lead to an undesired migration and therefore spreading of pollutants. Otherwise, contaminant volatilization is an effective thermal remediation process (Davis, 1997; McGee, 2003; Heron et al., 2005; Hiester and Bieber, 2017). A key parameter for the volatilization of organic pollutants consists in the equilibrium partitioning between gas phase and aqueous phase and is expressed by the Henry's law constant (H) (Heron et al., 1998). For the experimental determination of H several methods have been developed (Mackay and Shiu, 1981; Lincoff and Gossett, 1984; Gossett, 1987). For example, H can be calculated using independently measured solubilities and vapour pressures. Another method is to directly measure the concentration of a substance in the aqueous phase and the gas phase at equilibrium. Furthermore, H can be determined by measuring relative concentration changes within one phase during an equilibrium air-water exchange process (Batch air stripping) (Mackay and Shiu, 1981). Currently, one of the most popular methods is the EPICS technique (Equilibrium Partitioning In Closed Systems), which was introduced by Lincoff and Gossett (1984) and refined by Gossett (1987). For low- H compounds, where the determination of H via gas phase might fail, alternative methods have been developed, using dissolved concentrations at an EPICS-similar method (Kim and Kim, 2014), a kinetic method based on solute volatilization and air-phase transfer coefficients (Chao et al., 2017) or using experimentally determined thermodynamic descriptors (Abraham et al., 2019). If no experimental data are available for a compound of interest or if experimental data cannot be obtained (e.g. due to availability or toxicity of the compound), H and its temperature dependence can be modelled using the quantitative structure-activity relationship model (QSAR) (Nirmalakhdan and Speece, 1988; Nirmalakhdan et al., 1997). However, the results of these models often reveal significant deviations from experimental data.

Several studies have already been published focusing on the temperature dependence of H of organic pollutants (e.g. Ashworth et al., 1988; Gossett, 1987) and have been compiled by Staudinger and Roberts (2001) and Warneck (2007). Both compilations use a two-parameter fitting equation, which was considered as insufficient for larger temperature intervals as the changes in heat of dissolution is not considered (Heron et al., 1998). Furthermore, the works of Staudinger and Roberts (2001) and Warneck (2007) include H predominantly below 50 °C and 70 °C for few compounds, respectively. More recent studies, some of which dealt with temperatures up to 93 °C (e.g. Chen et al., 2012; Knauss et al., 2001; Görgenyi et al., 2002), were not included in existing compilations. Thus, a new compilation of available H for 41 pollutants considered in this study (Table 1), would increase the amount of data between 0 and 95 °C by 60% and between 40 and 95 °C even by 160%. In addition, for 14 out of 41 pollutants the temperature dependency of H could be expressed up to above 90 °C instead of up to 40–70 °C. A current compilation with appropriate data fitting is therefore necessary for the evaluation of the database regarding H

at elevated temperatures to spot existing data gaps. Although numerous studies on the temperature dependence of H already exist, the availability of data for higher temperatures are limited. Even for a comparatively widely investigated compound such as for example Tetrachloroethylene (PCE) there are over 60 data points up to 55 °C but only 8 between 55 and 95 °C, which moreover are partly contradictory. For a less commonly studied compound example Ethylbenzene there are only two data points above 55 °C with a maximum temperature of 70 °C or for 1,1-Dichloroethylene (1,1-DCE) no data above 40 °C could be found. Accordingly, further measurements at temperatures above 40 °C are necessary for many regarded compounds.

The aim of this study therefore is, firstly, to recompile available H in the literature with regard to data quality and quantity, especially at temperatures between 40 °C and 100 °C, and secondly, to reduce existing data gaps by additional measurements between 10 and 90 °C and to provide an updated compilation for H in the temperature range of 0–95 °C for 41 organic contaminants including new measurements.

2. Theoretical background

The dimensionless H described as the volatility constant is defined as the relationship between concentration of the compound in the gaseous (C_g) and the aqueous phase (C_w) (Schwarzenbach et al., 1993):

$$H = \frac{C_g}{C_w} \quad (1)$$

The other commonly used expression of H has the unit of [atm m³ mol⁻¹] and is usually expressed as k_H via the vapour pressure (P_v) (Heron et al., 1998; Schwarzenbach et al., 1993):

$$k_H = \frac{P_v}{C_w} \quad (2)$$

H and k_H relate to the temperature (T) and ideal gas constant (R) (Heron et al., 1998):

$$k_H = H R T \quad (3)$$

In terms of thermodynamics, H is directly linked to the fugacity f_i and activity coefficient γ_i with the mole fraction x_i of a compound (Munz and Roberts, 1986):

$$k_H = \frac{f_i}{x_i \gamma_i} \quad (4)$$

Accordingly, anything that changes the activity coefficient of a compound in solution will also affect its H . Studies have already shown that compared to a deionized water, H of a dissolved VOC increases by at least 10% when the ionic strength exceeds 0.2 M KCl or NaCl (Gossett, 1987; Peng and Wan, 1998). But also cosolvents, which are commonly used for methodological convenience when measuring H , influence the activity coefficients and thus H . However, the extent of influence essentially depends on the concentration of the cosolvents present and only becomes relevant at high cosolvent concentrations (Munz and Roberts, 1986). For example, the activity coefficient of chloroform was reduced by 10, 22 and 40% (10, 25, 50% for carbon tetrachloride) under the presence of methanol with a mole fraction of 0.02, 0.05 and 0.1, respectively, in the solution, although a reduction of the activity coefficient was only detectable above a methanol mole fraction of 0.005.

To obtain the temperature dependency of H , it has to be measured at different temperatures and the experimental data are correlated. For correlation of larger temperature intervals (e.g.

Table 1

Summary of temperature dependence of Henry's law constants for different compounds expressed by the regression parameters A, B, C in the given temperature range, R^2 = coefficient of determination, n = number of included data).

Compound	T-range [°C]	Reference (No. from footnote)	Regression parameter				
			$\ln(H) = A - B/T + C \ln(T)$ with H [-] and T [K]				
			A	B	C	R^2	n
Chlorinated hydrocarbons							
Chloromethane (<i>Methyl Chloride</i>)	0 - 93	1, 2, 8, 15, 45	164.514	9538.1	-23.426	0.958	27
Dichloromethane	5 - 93	5, 8, 9, 13, 39, 45	137.310	8923.3	-19.246	0.961	42
Trichloromethane (<i>Chloroform</i>)	2 - 93	4, 5, 6, 8, 9, 11, 18, 32, 39, 45	164.981	10720.0	-22.979	0.933	75
Tetrachloromethane (<i>Carbon Tetrachloride</i>)	2 - 93	4, 8, 9, 11, 13, 14, 18, 19, 39, 45	373.715	19941.3	-53.821	0.961	62
Chloroethane (<i>Ethyl Chloride</i>)	8 - 93	8, 9, 45	212.454	11752.3	-30.493	0.979	15
1,1-Dichloroethane	2 - 93	8, 9, 10, 13, 14, 18, 21, 32, 45	232.792	13489.4	-33.172	0.973	46
1,2-Dichloroethane	2 - 93	9, 10, 13, 14, 18, 45	367.459	19937.2	-53.270	0.868	26
1,1,1-Trichloroethane	2 - 93	4, 5, 8, 9, 13, 14, 17, 18, 19, 28, 45	459.803	23465.6	-66.958	0.922	57
1,1,2-Trichloroethane	2 - 50	9, 13, 14, 17, 18	196.111	12330.0	-27.737	0.974	19
	2 - 90	Combined with data measured in this study	111.318	8598.1	-15.053	0.987	23
1,1,1,2-Tetrachloroethane	20 - 40	13, 14	-627.459	-24549.8	95.262	0.942	7
	10 - 90	Combined with data measured in this study	643.414	34195.7	-93.240	0.957	15
1,1,2,2-Tetrachloroethane	10 - 40	9, 13, 14	-1526.119	-64857.6	228.960	0.586	12
Vinyl chloride	8 - 91	8, 9, 45	-17.713	1196.2	3.813	0.931	15
	8 - 91	Combined with data measured in this study	-16.658	1145.5	3.597	0.935	20
1,1-Dichloroethylene	2 - 40	8, 9, 13, 25	357.062	18641.4	-51.677	0.979	22
	2 - 90	Combined with data measured in this study	272.503	14818.7	-39.091	0.990	31
1,2-cis-Dichloroethylene	1.8 - 91	8, 9, 13, 14, 35, 45	39.763	4457.4	-4.664	0.972	27
1,2-trans-Dichloroethylene	1.8 - 70	8, 9, 13, 14, 19, 35	49.275	5027.6	-5.850	0.984	24
	1.8 - 90	Combined with data measured in this study	177.760	10748.7	-25.030	0.983	28
Trichloroethylene	1.8 - 95	5, 8, 9, 11, 13, 14, 17, 18, 22, 23, 25, 27, 28, 29, 32, 35, 45	88.308	7242.0	-11.411	0.973	106
	1.8 - 95	Combined with data measured in this study	110.411	8218.0	-14.715	0.973	115
Tetrachloroethylene	1.8 - 91	5, 8, 9, 11, 17, 18, 22, 27, 28, 35, 45	269.309	15787.9	-38.035	0.967	69
	1.8 - 91	Combined with data measured in this study	306.515	17408.7	-43.611	0.968	78
1,2-Dichloropropane	2 - 70	9, 13, 14, 25, 32	177.116	11180.7	-24.893	0.976	21
Monoaromatic hydrocarbons							
Benzene	0 - 89	3, 9, 17, 18, 20, 22, 29, 32, 34, 42, 44, 47	246.963	14043.0	-35.340	0.984	56
Ethylbenzene	2 - 70	9, 17, 18, 41, 42	571.229	29076.8	-83.347	0.984	20
	2 - 90	Combined with data measured in this study	220.665	-13454.3	-31.024	0.975	29
Propylbenzene	10 - 30	9	-103.290	-1594.1	17.050	0.997	5
Chlorobenzene	2 - 70	9, 25, 32, 44	322.326	17338.1	-46.683	0.953	18
	2 - 90	Combined with data measured in this study	190.288	11451.5	-45.382	0.947	27
1,2-Dichlorobenzene (<i>o</i> -Dichlorobenzene)	10 - 30	9	-2848.646	-123827.5	426.617	0.790	5
1,3-Dichlorobenzene (<i>m</i> -Dichlorobenzene)	10 - 30	9	-1475.902	-62683.5	221.782	0.884	5
1,4-Dichlorobenzene (<i>p</i> -Dichlorobenzene)	10 - 30	9	-1396.419	-59065.8	209.956	0.990	5
1,2,4-Trichlorobenzene	2 - 30	9, 25	-691.610	-28320.2	104.319	0.232	10
Pentachlorobenzene	14.8 - 50.5	12	192.339	13042.3	-26.701	0.991	6
Toluene	0 - 89	3, 9, 10, 17, 18, 22, 25, 29, 32, 34, 40, 41, 42, 44, 47	322.281	17633.4	-46.425	0.970	70
2,4-Dichlorotoluene	7.8 - 50	46	51.762	6354.8	-5.677	0.994	4
p-Xylene	0 - 50	3, 9, 16, 17, 18, 40, 47	690.635	34608.8	-101.065	0.917	26
	0 - 90	Combined with data measured in this study	325.315	18501.7	-46.436	0.946	30
o-Xylene	0 - 70	3, 9, 17, 18, 41, 42, 47	565.741	28883.6	-82.576	0.936	26
m-Xylene	0 - 70	9, 17, 18, 47	432.905	22590.2	-62.896	0.956	16
Polyaromatic Hydrocarbons							
Anthracene	4.1 - 35.3	20, 26	-203.722	-4966.5	31.759	0.879	12
Phenanthrene	4.1 - 34.7	20, 24	447.075	23724.2	-65.603	0.881	12
Pyrene	4.1 - 31	24	7.922	5078.8	0.321	1.000	5
Naphthalene	2 - 35.5	20, 25	-551.765	-20877.5	83.860	0.533	14
Acenaphthene	4.1 - 31	24	30.841	6909.0	-2.205	1.000	5
Fluoranthene	4.1 - 50.5	12, 24	38.789	6855.4	-4.047	0.695	11
Other							
2,4,5-Trichlorobiphenyl	4 - 31	26	-1723.949	-71259.7	259.866	0.912	5
MTBE	3 - 70	17, 29, 30, 31, 36, 38, 41, 42, 43	258.241	15697.9	-36.674	0.920	47
Phenol	4 - 29	7, 33, 37	1069.045	50360.9	-159.859	0.273	19

¹Glew & Moelwyn-Hughes (1953); ²Boggs & Buck (1958); ³Wasik & Tsang (1970); ⁴Hunter-Smith et al. (1983); ⁵Lincoff & Gossett (1984); ⁶Nicholson et al. (1984); ⁷Abd-El-Bary et al. (1986); ⁸Gossett (1987); ⁹Ashworth et al. (1988); ¹⁰Lamarche & Droste (1989); ¹¹Tancrede & Yanagisawa (1990); ¹²Hulscher et al. (1992); ¹³Tse et al. (1992); ¹⁴Wright et al. (1992); ¹⁵Elliott and Rowland (1993); ¹⁶Hansen et al. (1993); ¹⁷Robbins et al. (1993); ¹⁸Dewulf et al. (1995); ¹⁹Hansen et al. (1995); ²⁰Alaee et al. (1996); ²¹Hovorka & Dohnal (1997); ²²Peng & Wan (1997); ²³Heron et al. (1998); ²⁴Bamford et al. (1999)*; ²⁵Dewulf et al. (1999); ²⁶Bamford et al. (2000)*; ²⁷Knauss et al. (2000)**; ²⁸Vane & Giroux (2000); ²⁹Bierwagen & Keller (2001); ³⁰Callender & Davis (2001); ³¹Fukuchi et al. (2002); ³²Görgenyi (2002); ³³Harrison et al. (2002); ³⁴Bakierowska & Trzeczczynski (2003); ³⁵Shimotori & Arnold (2003); ³⁶Arp & Schmidt (2004); ³⁷Feigenbrugel et al. (2004); ³⁸Fischer et al. (2004); ³⁹Lutsyk et al. (2005); ⁴⁰Lin & Chou (2006); ⁴¹Falabella (2007); ⁴²Sieg et al. (2009); ⁴³Hwang et al. (2010); ⁴⁴Lau et al. (2010); ⁴⁵Chen et al. (2012); ⁴⁶Brockbank et al. (2013); ⁴⁷Kim & Kim (2014).

*The authors displayed H predicted by linear regression based on their experimental data; **The authors calculated H using self-measured solubilities at the respective temperature.

between 10 and 90 °C) it is important to consider the changes in the heat of dissolution that is accomplished by using the 3-parameter (A, B, C) equation which was derived by [Heron et al. \(1998\)](#):

$$\ln k_H = A - \frac{B}{T} + C \ln T \quad (5)$$

3. Methods

3.1. Compilation of published Henry's law data

To check the availability and quality of H in the temperature range between 0 and 100 °C, 47 publications concerning temperature dependence of H were found and H of 41 compounds including chlorinated, mono-, polyaromatic hydrocarbons and other were compiled ([Table 1](#)). Thereby only H were considered which were examined in deionized water, since differences in ionic strength result in differing activity coefficients and thus H . Conversely, a different use of cosolvents was not considered, as the concentrations of used cosolvents were too low to exert a significant influence on activity coefficients ([Gossett, 1985](#); [Gossett et al., 1985](#); [Munz and Roberts, 1986](#)). As H are presented in various forms and units (e.g. [Sander, 2015](#)), the collected data were unified and are presented here as a dimensionless ratio between concentration of a compound in the gaseous and aqueous phase (Eq. (1)).

To describe the temperature dependence of H , the 3-parameter regression equation by [Heron et al. \(1998\)](#) (Eq. (5)) was chosen, whereas the calculation of the regression parameters A, B, C was performed for each investigated substance over all collected data using the Linest-Function in Microsoft Excel (2019) (64 bit edition).

3.2. Measuring of Henry's law constants

3.2.1. Chemicals and stock solutions

10 compounds were selected ([Table 2](#)) which are frequently found as groundwater contaminants and for which the literature review revealed quantitative data deficits at temperatures above 40 °C or widely differing H (e.g. factor 2 at a certain temperature). Although interactions between cosolvents and an influence on

activity coefficients cannot be completely excluded (see section 2), the methodology in this study was oriented on previous research to provide comparability and methodological convenience. To ensure constant compound concentrations during every measurement series, stock solutions of the compounds were prepared with methanol as solvent for the examined compounds. For 1,1,2-Trichloroethane (1,1,2-TCA) (99.7% - LGC, Germany), 1,1,1,2-Tetrachloroethane (1,1,1,2-PCA) (99.8% - LGC, Germany), Vinyl chloride (VC) (5000 mg/L - LGC, SC USA), 1,2-trans-Dichloroethylene (1,2-trans-DCE) (99.9% - Merck, Germany), Ethylbenzene, Chlorobenzene, and p-Xylene (98% - Restek, Germany) a separate stock solution was prepared for each compound consisting solely of 20 mL methanol (99.7% - Merck, Germany) and 0.13–0.29 mL of the compound (exact amounts are listed in [Table A1](#)). The added volume of the compound was varied for analytical reasons to provide a clearly evaluable peak area for each compound analyzed with gas chromatography (GC). Another stock solution was prepared containing 0.10–0.13 mL 1,2-Dichloroethane (1,2-DCA) (98% - Merck, Germany), 1,1-Dichloroethylene (1,1-DCE) (98% - Merck, Germany), Trichloroethylene (TCE) (98% - Sigma-Aldrich, Germany) and Tetrachloroethylene (PCE) (95% - Merck, Germany) and 20 mL Methanol.

3.2.2. EPICS-procedure

The experiments to determine H of the selected compounds were performed with the EPICS method after [Gossett \(1987\)](#) at temperatures between 10 and 90 °C. For each substance at each investigated temperature, the experiment was carried out at first in 4 and later in 3 pairs consisting of 120 mL glass vials, half of which were filled with 20 mL and the other half with 80 mL of deionized water. To prevent an increase in pressure inside the vial caused by thermal expansion of the water, the water and the vials were pre-heated to the experimental temperature before the water was filled into the vials. Immediately after filling the vials, they were sealed with Teflon coated septa and aluminum crimp caps (Macherey & Nagel, Germany). 100 µL of the previously prepared stock solution was then added to each vial by injecting it through the septum with a 100 µL syringe. To eliminate inaccuracies, the exact added mass of the stock solution was determined gravimetrically. The vials were then submerged upside down in a tempered water bath for

Table 2

Dimensionless Henry's law constants measured in this study with percentual standard deviations (%SD) and calculated regression parameter A, B, C with respective coefficients of determination (R²).

t [°C]		1,1,2-TCA	1,1,1,2-PCA	VC	1,1-DCE	1,2-trans-DCE	TCE	PCE	Ethyl-benzene	Chloro-benzene	p-Xylene
10	H [–]	–	0.01	–	0.57	–	0.19	0.31	0.15	0.1	–
	SD%	–	25.7	–	8.6	–	10.1	8.18	9.01	15.4	–
20	H [–]	–	0.05	–	0.81	–	0.31	0.53	0.24	0.17	–
	SD%	–	33.8	–	5.83	–	8.25	7.58	10.3	11.9	–
30	H [–]	–	–	–	1.27	–	0.51	0.91	0.44	0.24	–
	SD%	–	–	–	7.24	–	7.05	8.12	4.88	6.31	–
40	H [–]	0.08	0.22	1.43	1.64	0.64	0.72	1.27	0.59	0.28	0.58
	SD%	4.9	9.17	2.64	9.95	6.51	7.04	6.58	6.57	17.7	7.92
50	H [–]	–	0.28	–	2.08	–	0.96	1.76	0.81	0.47	–
	SD%	–	4.95	–	7.42	–	4.33	5.71	7.85	8.55	–
60	H [–]	0.14	0.37	2.47	2.43	0.99	1.24	2.23	1.05	0.53	0.99
	SD%	10.2	11.1	8.93	11.2	1.84	8.35	13.8	8.49	6.11	4.19
70	H [–]	–	0.42	2.29	2.83	–	1.49	2.64	1.45	0.63	–
	SD%	–	10.9	3.13	15.4	–	11	15.4	10.6	8.99	–
80	H [–]	0.23	0.75	2.5	3.37	1.54	1.88	3.32	1.97	0.72	1.45
	SD%	9.36	7.1	3.33	17	12.4	10.7	18	14.4	3.23	5.99
90	H [–]	0.37	0.77	4.52	3.72	1.73	2.08	3.7	2.67	0.92	2.32
	SD%	16.6	12.8	6.28	18.7	5.16	13.3	23.1	27.5	11.2	7.42
A		–295.178	650.032	–314.560	217.685	66.774	249.668	310.103	107.956	710.418	–305.829
B		–11,638.3	34898.2	–13,701.6	12299.0	5275.5	14406.5	17321.9	8193.7	36201.3	12,575.3
C		44.461	–94.026	47.195	–30.963	–8.767	–35.500	–44.295	–14.329	–103.629	46.139
R²		0.992	0.987	0.837	0.998	0.997	0.999	0.999	0.996	0.973	0.982

approximately 48 h at the desired temperature, which was sufficient to reach equilibria between the gaseous and aqueous phase, according to likewise studies that have revealed the equilibrium to be reached within a few hours for several compounds (Gossett, 1985; Heron et al., 1998).

Afterwards, 1 mL gas samples were taken from the headspace of the vials using a gas-tight syringe (Model 1001 gas tight, Hamilton - USA) with a pressure valve (SLLV, SGE - Australia) and immediately released into 20 mL GC glass vials which were previously sealed with PTFE coated septa and aluminum crimp caps. For each EPICS vial the procedure was performed three times to mitigate measurement uncertainties. Although a decrease in measured peak area of 3% ($\pm 9\%$) on average between first and third sample was observed, the resulting error was considered to be less influential than the error due to sampling procedure and analytical uncertainties. During sampling, the EPICS vials were kept under water to avoid temperature changes and thus condensation inside the vials. Also, no condensation could be observed inside the syringe at any temperature during sampling. The headspace concentrations of the compounds were measured by GC with a flame ionization detector (FID) detector (Hewlett Packard, 6890 series, column: AT-Q - Alltech for VC and column: HP-PLOT/Q - Agilent for the remaining compounds). Since no absolute concentrations are required for the calculation of H (Hansen et al., 1995), but only the ratio between the 40 mL and 100 mL headspace concentration is necessary, the GC did not have to be calibrated beforehand, as the precondition of a linear relationship between peak area and concentration was fulfilled within the expected concentration range. Consequently, the peak areas were sufficient for the calculation of H .

Using the average value of the three measured peak areas for each EPICS vial and the mass of the added stock solution, dimensionless H were calculated using a modification of the Gossett (1987) equation, where GC peak areas are used instead of concentrations:

$$H = \frac{V_{w2}A_2M_1 - V_{w1}A_1M_2}{V_{g1}A_1M_2 - V_{g2}A_2M_1} \quad (6)$$

where $V_{w\#}$ and $V_{g\#}$ represent the volumes of water and gas in EPICS vials, $M_{\#}$ refers to the mass of solute added to the vials and $A_{\#}$ is the GC peak area of volatile compounds in the gas phase. The subscripts 1 and 2 describe the EPICS vials with different volumes of water. When using several pairs of vials, each vial with a lower water volume can be related with each vial with a higher water volume, resulting in a total of 9 H when using 3 pairs of EPICS vials and 16 when using 4 pairs. Out of these 9 or 16 H the mean value was determined and additionally the standard deviation was calculated. To describe the temperature dependency, Eq. (5) was used, whereby the respective mean value of H was taken.

4. Results and discussion

4.1. Compilation of published Henry's law constants

4.1.1. Availability of Henry's law constants at elevated temperatures

In general, the review of 47 publications on the temperature dependency of H of 41 organic pollutants revealed a large amount of data available in the temperature range between 10 and 40 °C, but the amount of data decreases significantly at higher temperatures (Fig. 1). For 15 out of 41 reviewed compounds, there were no H at temperatures above 40 °C available and for another 12 compounds, the highest examined temperature was between 50 and 70 °C (Table 1; Fig. S1-S4). Although there are 14 compounds with H up to 90 °C, however, the results around 90 °C can scatter between different measurements by a factor of 2 or more (e.g. for TCE and

PCE, Fig. 2) which would lead to significant uncertainties in calculations on gas phase formation. Although an increase in H with increasing temperature is expected, a doubling of the value within 1 °C is unlikely. Such discontinuities of H at small temperature differences occurred for almost all substances Chen et al. (2012) investigated (Fig. S1-S4).

Currently there are only three publications (Heron et al., 1998; Lau et al., 2010; Chen et al., 2012) that have investigated H at temperatures up to 90 °C for the compounds reviewed in this study. Heron et al. (1998) investigated solely TCE while Lau et al. (2010) examined Benzene and Toluene in this temperature range. Therefore, for the remaining 11 compounds with H investigated up to 90 °C only data from Chen et al. (2012) are available. Their data are consistent below 80 °C, but especially in the range of 90 °C for several compounds rapid increases of H within small temperature differences are evident which have not been verified by other studies so far. Accordingly, at temperatures above 70 °C it was apparent that measured H increasingly deviated from the calculated regression. Due to the small amount of data in this temperature range which often also scatter widely, the results of the regressions are not sufficiently reliable and further measurements of H above 70 °C were indispensable. However, large deviations in the data mainly occur at temperatures above 70 °C, whereas at lower temperatures there is usually a good agreement between the results of the different studies, despite the use of different methods to measure H . Thus, the regression calculated here fits the majority of the published data very well and provides a high precision.

4.1.2. Temperature dependency of H

The calculated regression parameters A , B and C for each of the examined compounds for the illustration of the temperature dependence of H are listed together with the coefficient of determination (R^2) in Table 1. Using these data, the dimensionless H in the given temperature range can be calculated by Eq. (5) with a chosen temperature in Kelvin (Fig. 2).

As expected, H showed a clear temperature dependence, increasing 5- to 18-fold between 10 and 90 °C for those compounds with available data up to 90 °C (Fig. 3). The effect of temperature on H appears to be independent of compound class, although based on current available data this statement cannot be conclusively determined, since for many compounds no or insufficient data exist at temperatures above 40 °C. This is especially valid for the polyaromatic hydrocarbons, where the slopes of the normalized curves appear to be comparatively steep between 20 and 40 °C (Fig. S5), indicating a stronger temperature dependency compared to other groups. However, the absolute values of H are indeed related to compound groups. Thus, it was found that on average the chlorinated ethylenes have the highest H (0.2–1 at 25 °C) while the aromatic hydrocarbons have usually lower H (<0.45 at 25 °C) and H are especially low for the polyaromatic hydrocarbons (<0.02 at 25 °C).

4.2. Measurements of Henry's law constant

4.2.1. Description of experimental results and comparison with literature data

As expected, all investigated compounds showed strictly monotone increasing H with rising temperature (Table 2, Fig. 4). The measurements revealed a 6.5- (1,1-DCE) to 51-fold (1,1,2-PCA) increase in H between 10 and 90 °C, which underlines the variation of temperature dependence of H of organic pollutants. For TCE and PCE, H increased approximately by a factor of 11 and 12, respectively, which is in good agreement with observations of Chen et al. (2012), who found an about 12- and 13-fold increase of H between 8 and 90 °C for TCE and PCE, respectively. At temperatures

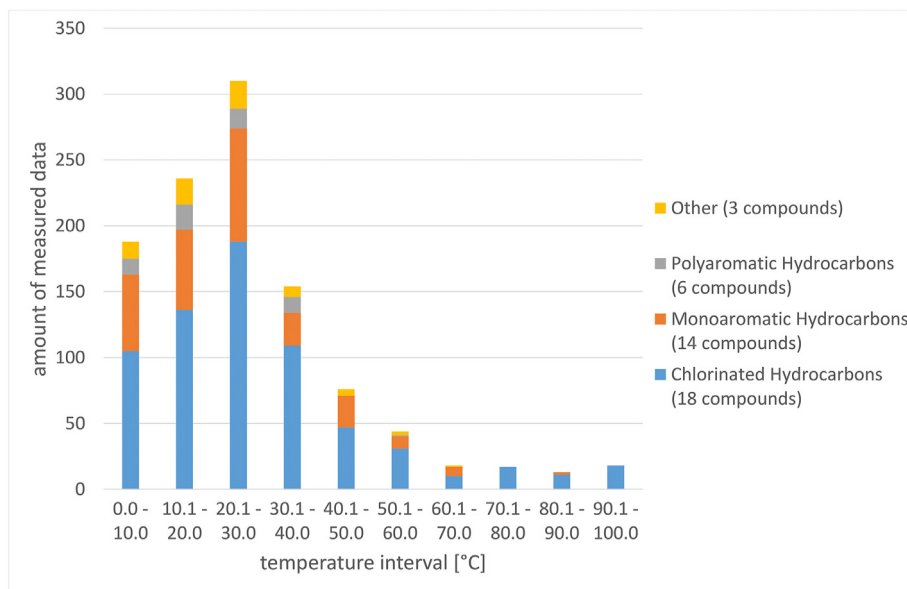


Fig. 1. Cumulative occurrence of published Henry's law constant in defined temperature intervals.

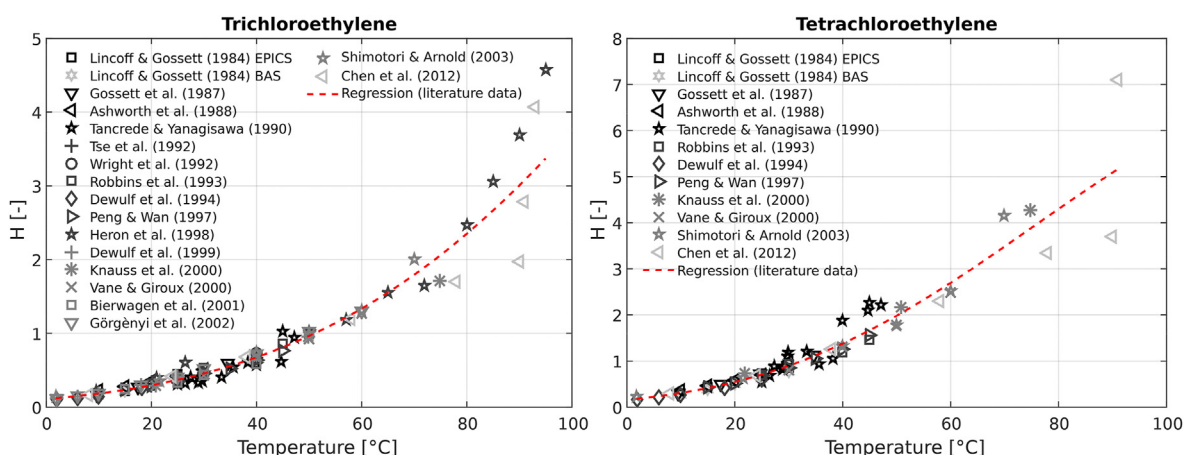


Fig. 2. Literature data of H for TCE and PCE with calculated regression curves, compiled data for the remaining 39 compounds are shown in Fig. S1-S4.

up to 40 °C, a very high correlation between H measured in this study and the results of other publications was achieved for all substances examined (for VC, 1,2-trans-DCE, TCE and PCE even up to 60 °C) (Fig. 4). Thus, the comparability of the measurements of this study with those previous studies is given and thereby support the results of this study. Overall, the measurements provided acceptable standard deviations (Table 2) and high R^2 (0.837–0.999). Nevertheless, some particularities were found in our results, for example, for 1,1,1,2-PCA H increases with a relatively high leap between 70 and 80 °C from 0.42 ± 0.05 to 0.75 ± 0.05 (Fig. 4). Such a remarkable increase seems unusual in relation to the general slope of the curve but was accepted as the standard deviations were acceptable. For VC, H was not increasing continuously with rising temperature, as H was higher at 60 °C than at 70 and 80 °C. Chen et al. (2012) also measured a likewise discontinuous increase in H with rising temperature but in both studies the respective standard deviations were low, thus the individual results were not questioned. For the other investigated compounds, in some cases differences between the results of this study and those of other studies occurred, which could be caused by

methodological variations (presence of cosolvents, measures for pressure equalization), however, no clear systematics were found.

The measurements of H presented in this study demonstrated on the one hand a good agreement with the results of other publications especially at temperatures up to 40 °C. On the other hand, our measurements completed the data set on H for the investigated compounds at temperatures between 40 and 90 °C, since data were collected for the first time for many compound-temperature combinations in this temperature range. The standard deviations of the experiments in this study were between 2.6 and 35% but mainly in a single digit percentage (Table 2) and thus in a method-typical range, in which also the results of other authors (e.g. Ashworth et al., 1988; Heron et al., 1998; Chen et al., 2012) were found. Overall, it is noticeable that the standard deviations increased with increasing temperature and were also higher at low H (Table 2). Similar observations were also made by Heron et al. (1998) and Chen et al. (2012) indicating a flaw in the methodology which might cause uncertain results at higher temperatures or at low H . The hypothesis by Chen et al. (2012) according to which the applied EPICS-method is less suitable for H above 3, could not be confirmed

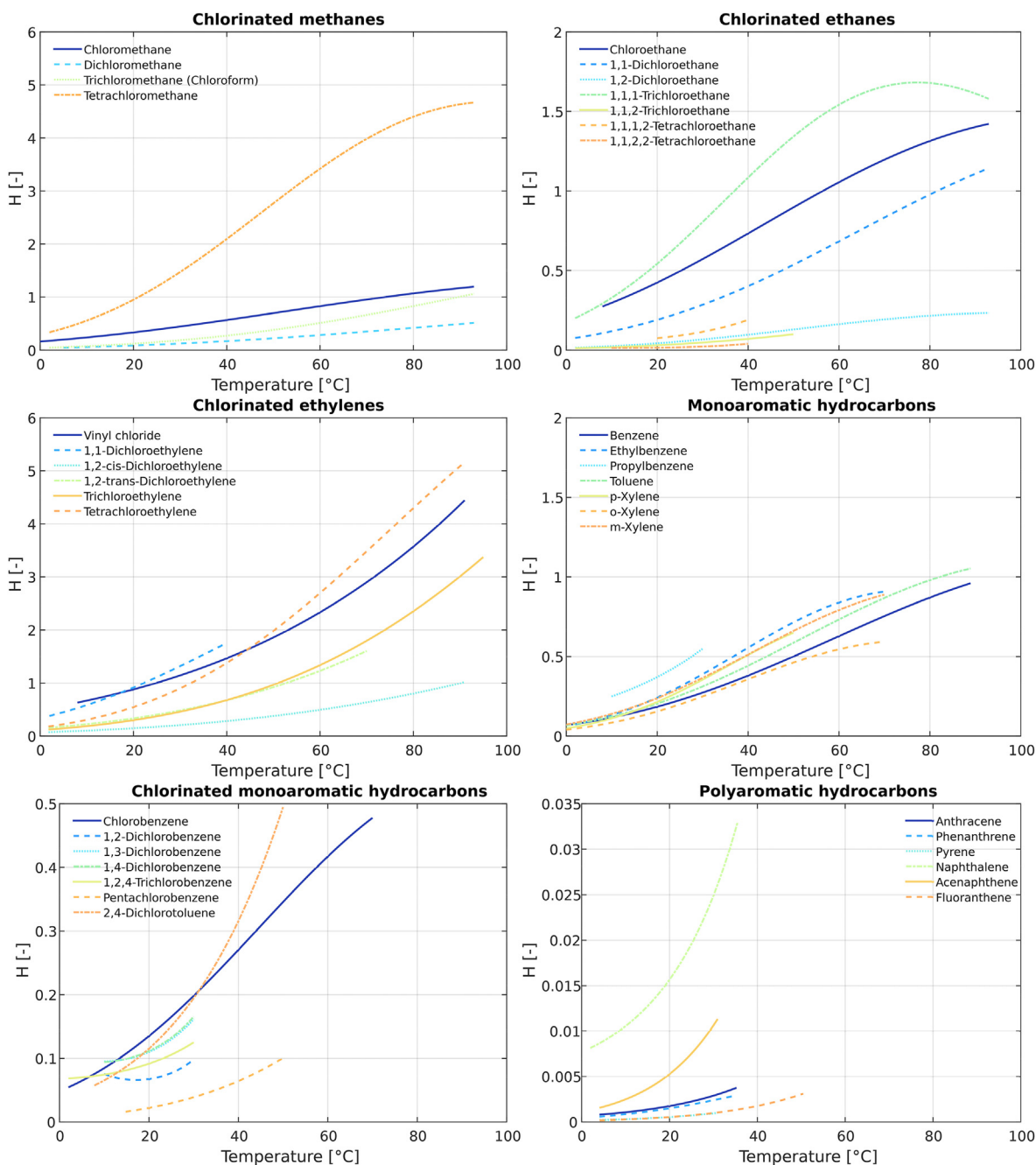


Fig. 3. Regression curves for reviewed compounds, based on literature data, expressed as dimensionless H . The underlying data for the regression curves are presented in Fig. S1-S4 and the respective R^2 in Table 1.

in this study, since in particular VC at 90 °C with a measured H of 4.52 and a corresponding standard deviation of 6.3% provided a very precise result. The standard deviations of the other compounds with H above 3 were approximately in the range typical for the respective temperature.

4.2.2. Methodological differences between studies concerning Henry's law constants at elevated temperatures

H at elevated temperatures (e.g. ≥ 50 °C) were mostly obtained by EPICS experiments or modified EPICS experiments. However, there was some variation between the considered studies that might lead to the observed different results.

One difference between the studies consisted in the use of

methanol as solvent for the VOCs, which was used in this study as well as in many other studies (e.g. Görgenyi et al., 2002; Shimotori and Arnold, 2003; Chen et al., 2012). The advantage of using methanol as solvent consists in its miscibility with VOCs, which ensures a homogeneous distribution of the VOCs in the stock solution, but in some studies water was used as solvent (e.g. Heron et al., 1998; Falabella, 2007) or pure compounds were added to the EPICS vials (e.g. Lau et al., 2010). Methanol was widely applied although the presence of a cosolvent such as methanol affects the activity coefficient and thus also H of the investigated compounds. However, the applied methanol concentrations (<0.5 vol%/mole fraction of $\sim 2 \times 10^{-3}$ in this study) are expected to be sufficiently low to avoid significant impacts. This assumption was confirmed on

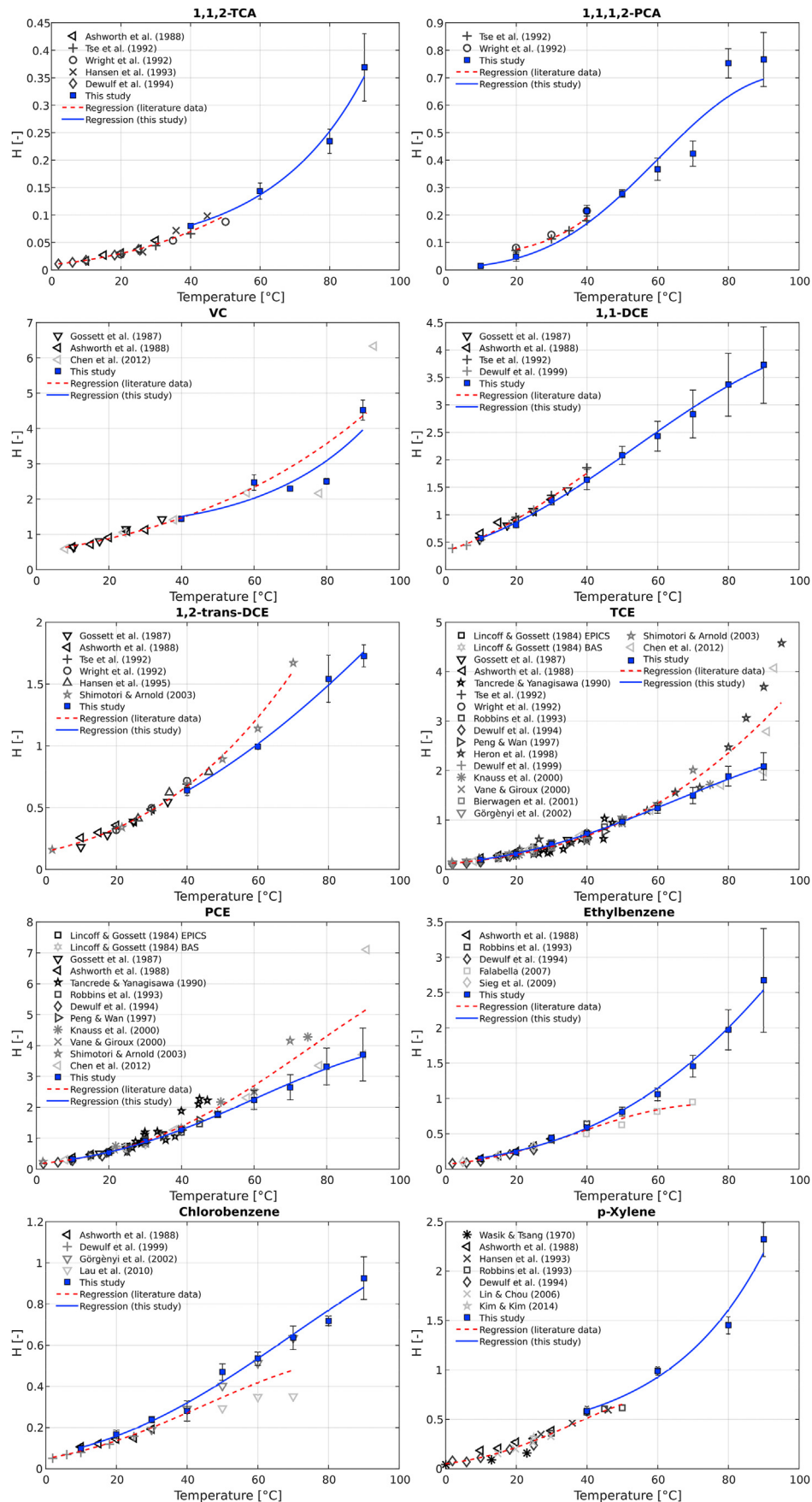


Fig. 4. Measured H of selected compounds plotted together with literature data. Vertical lines represent standard deviations.

the one hand by Munz and Roberts (1986), who observed a quantifiable influence on activity coefficients only above a methanol mole fraction of $>5 \times 10^{-3}$ and on the other hand by Gossett (1985), who directly investigated the influence of methanol on H and only observed a significant change in H at a volume fraction of more than 2% methanol. However, the experiments by Munz and Roberts (1986) and Gossett (1985) were only performed at room temperature, while the differences between the measured H among various studies only occurred at elevated temperatures. Consequently, currently an influence of methanol on activity coefficients and accordingly on H cannot be completely excluded at elevated temperatures. Since thermodynamic parameters are usually strongly temperature-dependent, it is not unreasonable that impacts of interactions between different compounds, e.g. changes of activity coefficients could also be temperature-dependant. But a significant influence is not necessarily to be expected since regardless of temperature no systematic trends of the results between studies with and without methanol were observed. To verify if methanol at low concentrations (<0.5 vol%) has no significant influence on H at elevated temperatures, exemplarily H of 1,2-trans-DCE and of p-Xylene were investigated at 60 and 80 °C with the same methodology as in previous experiments but without methanol as solvent and with lower compound concentrations because of their low solubility in water (Appendix D). No systematic difference was found between the results of the respective experiments since the differences in H between experiments with and without methanol were low and ranged within the measurement uncertainties (Fig. S6).

Another methodological difference between the numerous studies is the number of compounds examined simultaneously, with some studies examining the compounds individually (with or without methanol present) (e.g. Heron et al., 1998; Lau et al., 2010), while other studies used stock solutions containing several compounds (dissolved in water or methanol) (e.g. Görgenyi et al., 2002; Shimotori and Arnold, 2003; Falabella, 2007; Chen et al., 2012). Also, in this case, other present compounds act as cosolvents similar to methanol, hence the discussion regarding the influence of cosolvents on activity coefficients also is valid here, but investigations were only conducted at moderate temperatures <30 °C, where no clear interactions could be detected when examining several compounds simultaneously (Gossett et al., 1985). Although it should be assumed that in addition to the concentration of a cosolvent the influence on activity coefficients also depends on the type of cosolvent. Those studies examining mixtures of several compounds at higher temperatures did not detect any irregularities in their results that would suggest interactions between the compounds (e.g. Görgenyi et al., 2002; Shimotori and Arnold, 2003; Chen et al., 2012), therefore changes in activity coefficients seem to be negligible for the compounds and concentrations used here.

Among the underlying studies, which performed the EPICS procedure at temperatures above 50 °C (e.g. Heron et al., 1998; Görgenyi et al., 2002; Shimotori and Arnold, 2003; Chen et al., 2012) or a modified EPICS technique (Falabella, 2007; Lau et al., 2010), none provided for a complete pressure equalization inside of the experimental vials. Only Görgenyi et al. (2002), Chen et al. (2012) and this study equalized the pressure increase caused by thermal expansion and outgassing of water vapour and dissolved atmospheric gases such as N_2 or O_2 . Although a moderate increase in pressure (by up to ~ 0.1 MPa, calculated with PHREEQC v3.6.2 (Parkhurst and Apello, 2013)) is still expected, since a part of the dissolved VOCs will be released to the gas phase according to Henry's law and no further pressure equalization was performed, however, the increase in pressure will be lower than without any measures. An influence of a moderate pressure increase on H is

currently not exactly known, only in high-pressure experiments (>10 MPa) a reduction of H due to increasing pressure was observed for different compounds (Sedlbauer et al., 2002; Majer et al., 2008; Lee et al., 2017). A complete pressure equalization is impossible with the common EPICS method, since an unknown part of the added VOC-mass would be released when releasing gas for pressure equalization, thus invalidating the calculation of H , which is based on mass balance. This might be a weakness of this method if any pressure differences in the range of e.g. 0.1 MPa are found to influence H . Presumably, the increased variance of the measured H at higher temperatures observed in this study as well as by Chen et al. (2012) and Heron et al. (1998) could be partly attributable to pressure increase inside the vials.

4.2.3. Combination of literature data with own measured data

Summarising, it can be stated that a few differences did occur between the results of our measurements and those of other studies, which might be of methodological origin. However, the differences did not appear to be systematic. Consequently, no data from previous researchers were excluded, hence, our measurements were added to calculate regression parameters based on all collected literature data and our results (Table 1), in addition to the regression curves based either on literature data or on our results shown in Figs. 3 and 4.

The combined regression curve well reflected both the H found in the literature as well as the measured H in this study (Fig. S7) and extends the temperature range covered by the regression curve which is now secured by measurements. Significant differences were noticeable for TCE as the combined regression curve is much closer to the regression curve based on literature data than to our data. This can be explained primarily due to significantly higher H measured by Heron et al. (1998) between 80 and 95 °C with a high data density (4 measurements), whereas only 2 measurements from this study were included in the regression calculation. Consequently, the results of Heron et al. (1998) are weighted higher in the combined regression compared to present data. Also, for PCE a difference between the combined regression curve and the one based on present data was apparent. In this case, the differences resulted in the comparably high H of 7.1 at 91 °C measured by Chen et al. (2012) compared to 3.7 at 90 °C measured in this study and by Chen et al. (2012), which significantly increased the regression curves.

5. Conclusions

The compilation of H provided an overview on the availability of H at elevated temperatures, which, however, also revealed data deficiencies even for common compounds. Based on the experiments on the temperature dependence of H in this study, we were able to close some of the existing data gaps for the most of the relevant compounds, thus completing the knowledge of compound behaviour due to temperature increases by e.g. UTES, conventional thermal remediation methods or UHI.

Referring to published H and those measured in this study, a clear increase in H (~ 5 – 50 -fold) between 10 and 90 °C can be observed for all compounds considered. Thus, a temperature increase e.g. by UTES would lead to a significant reduction of the dissolved mass of pollutants, since the fraction of the pollutant in the gas phase, which should be extracted to prevent a spread of contamination, would be increased. Even a moderate temperature increase from groundwater-typical temperatures such as 10 °C up to 40 °C would lead to a considerable reduction in the mass of dissolved pollutants, since the H in this temperature range are still at least doubled to quadrupled for the compounds investigated here. Such temperature increases of the subsurface are common at

UTES sites but can also be reached through infrastructure in urban areas (e.g. asphalt surfaces, subway tunnels etc.) so that even UHI could have an effect in the case of contaminations that should be taken into account.

However, except in the case of monoaromatic hydrocarbons, where the influence of rising temperature tends to be quite similar for each compound (~8-fold increase in H between 10 and 90 °C) (Fig. S5), the temperature dependence of H appears to be independent of the pollutant classification, since on the one hand no systematic differences between classes could be detected (Fig. S5) and on the other hand substantial differences also exist between individual compounds within a class (e.g. 1,1,1,2-tetrachloroethane and 1,1,1-trichloroethane with a 50-fold and 5-fold increase in H between 10 and 90 °C, respectively (Fig. 4; Fig. S5)). Therefore, compound classes cannot be generalised, and each compound must be viewed individually. But in terms of remediation potential, the increase of H with temperature is not the only relevant factor, since despite a high increase in H with rising temperature, H can still be low, keeping the main part of the pollutant mass in solution. This applies in particular to polyaromatic hydrocarbons, which have H in the range of 10^{-3} to 10^{-2} between 10 and 40 °C. In contrast, for chlorinated ethylenes, a temperature increase up to e.g. 70 °C which can easily be reached by (HT-)UTES would result in a significant remediation effect, since H then rises to well above 1 for most compounds, therefore leading to higher pollutant concentrations in the gas phase compared to the aqueous phase.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2021.129858>.

Credit author statement

Alexander Schwardt: Conceptualization, Methodology, Validation, Investigation, Writing – original draft; Andreas Dahmke: Supervision, Funding acquisition, Project administration, Writing – review & editing, Validation; Ralf Köber: Conceptualization, Methodology, Supervision, Writing – review & editing, Validation

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Corrigendum to “Henry’s law constants of volatile organic compounds between 0 and 95 °C – data compilation and complementation in context of urban temperature increases of the subsurface” [Chem. 272 (2021) 129858]

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The authors regret that the given regression parameters in Table 1 “Summary of temperature dependence of Henry’s law constants [...]” for ethylbenzene and chlorobenzene and in Table 2 “Dimensionless

Henry’s law constants measured in this study [...]” for chlorobenzene and p-xylene are erroneous. The correct values are listed below.

The authors would like to apologise for any inconvenience caused.

Table 1

Summary of temperature dependence of Henry’s law constants for different compounds expressed by the regression parameters A, B, C in the given temperature range, R^2 = coefficient of determination, n = number of included data).

Compound	T-range [°C]	Reference (No. from footnote)	Regression parameter				
			$\ln(H) = A - B/T + C \ln(T)$ with H [–] and T [K]				
			A	B	C	R^2	n
Ethylbenzene	2–70	9, 17, 18, 41, 42	571.229	29076.8	–83.347	0.984	20
	2–90	Combined with data measured in this study	220.665	13454.3	–31.024	0.975	29
Chlorobenzene	2–70	9, 25, 32, 44	322.326	17338.1	–46.683	0.953	18
	2–90	Combined with data measured in this study	190.288	11451.5	–26.972	0.954	27

⁹Ashworth et al. (1988). ¹⁷Robbins et al. (1993). ¹⁸Dewulf et al. (1995). ²⁵Dewulf et al. (1999). ³²Görgenyi (2002). ⁴¹Falabella (2007). ⁴²Sieg et al. (2009). ⁴⁴Lau et al. (2010).

Table 2

Dimensionless Henry’s law constants measured in this study with percentual standard deviations (%SD) and calculated regression parameter A, B, C with respective coefficients of determination (R^2).

	Chlorobenzene	p-Xylene
A	191.729	–305.829
B	11472.0	–12575.3
C	–27.188	46.139
R^2	0.991	0.982

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